
2019-Ph-H1-Q20

Section: Particles and Waves

Topic: Nuclear Reactions

Question:

What is conserved in a nuclear reaction?

Answer:

In nuclear reactions, **mass-energy**, **charge**, and **nucleon number** are conserved.

Correct Option: E (Mass-energy and nucleon number)

Guidance for Students:

Always check which conservation laws apply. Mass alone isn't conserved, but **mass-energy** is.

Revision Tips:

- Nucleon number = total of protons + neutrons
- Mass can convert to energy ($E = mc^2$), so **mass-energy** is conserved
- Charge is also conserved, but not asked here

Let me know when you're ready for **Questions 21–25!**

Here are **Questions 21–25** from the **2019 Physics Paper 1**, following our standard format with guidance and revision tips:
